



Prerequisite topics

- The prerequisites for this course on Particle Filters include all topics from the first three courses of the specialization:
 - Kalman Filter Boot Camp;
 - Linear Kalman Filter Deep Dive;
 - Nonlinear Kalman Filters.
- This lesson is a brief review (sometimes only an itemization) of the key topics from these courses, in case some time has elapsed since you studied them until now.
- Or, if you believe that you have learned the content of these courses from another source, this lesson will introduce the notation I use and also serve as a review of these important topics.
 - If you have a weak background in any topic, now would be a good time to review those concepts before proceeding much further in this course.



All Kalman filters use models

- All Kalman filters (and particle filters) require a model (i.e., a set of equations) that describes the system whose internal state we wish to estimate.
- This course assumes a nonlinear state-space model of the form:

$$x_{k+1} = f(x_k, u_k, w_k)$$

$$z_k = h(x_k, u_k, v_k).$$
- In this model, k is the discrete-time index and:
 - Variable x_k is the state (this is what we estimate), u_k is the input (which we measure), w_k is the (unmeasured) process noise, z_k is the output measurement (which we also measure), and v_k is (unmeasured) sensor/measurement noise.
 - All variables can be either vectors or scalars.
- Also, $f(\cdot)$ is the state/process equation; $h(\cdot)$ is the measurement/output equation.



Random variables

- Process noises w_k and sensor noises v_k are “random variables” (abbreviated as RVs).
 - We never measure their values directly, so the best we can do is to model some key attributes of the characteristics of these noises.
 - Because w_k is random, x_k is also an RV; because v_k is random, z_k is also an RV.
- RVs are most completely modeled using probability density functions (pdfs).
 - These pdfs can have infinite variety, but are always nonnegative and integrate to 1.
 - Until now, we have assumed that all pdfs are Gaussian; we will no longer do so.
- The expected value (mean) of an RV is denoted by $\mathbb{E}[\cdot]$ and variance by $\text{var}(\cdot)$.
 - Expected value is the center of mass of the pdf; variance is a measure of width.
 - Standard deviation is the square root of the variance.
 - Samples of an RV usually fall within a few standard deviations of the expected value (mean).



Joint random variables

- We often wish to find relationships between/among RVs.
 - If two RVs are uncorrelated, there is no linear equation that can help predict the value of one RV even if we are given the value of the other.
 - If two RVs are independent, there is no linear or nonlinear equation that can help predict the value of one RV given the value of the other.
- When considering multiple RVs together, we require joint pdfs; e.g., $f(x_k, z_k)$.
- We also require evaluation of conditional pdfs and conditional expectations.

$$f(x_k | z_k) = \frac{f(x_k, z_k)f(x_k)}{f(z_k)} \quad \text{and} \quad \mathbb{E}[x_k | z_k = z] = \int_{-\infty}^{\infty} x f(x | z) dx.$$
- When we consider a time-sequence of RVs (a “random process” or “stochastic process”), we also need to understand autocorrelation and cross-correlation.
 - Special example: A “white process” has zero mean and all time samples are uncorrelated with any other time sample in the process.



Kalman filters

- A Kalman filter (KF) seeks to use measurements of a system’s input sequence $\mathbb{U}_k = \{u_0, \dots, u_k\}$ and output sequence $\mathbb{Z}_k = \{z_0, \dots, z_k\}$, plus a system model, to estimate the system’s internal state x_k .
 - The estimate $\hat{x}_k$ is usually computed to be the conditional mean: $\hat{x}_k = \mathbb{E}[x_k | \mathbb{Z}_k]$, which is the “minimum mean-squared-error estimator” (MMSE).
- KFs use a predict/correct mechanism; we first predict the state: $\hat{x}_k^- = \mathbb{E}[x_k | \mathbb{Z}_{k-1}]$, and then update the prediction to make an estimate: $\hat{x}_k^+ = \mathbb{E}[x_k | \mathbb{Z}_k]$.
 - This allows for very efficient recursive implementation of the optimal MMSE estimator (assuming the assumptions made during the KF derivation are met).
- KFs also produce confidence bounds on the estimate via the estimation-error covariance matrix $\Sigma_{\hat{x},k}^+$, which is automatically computed as one of the KF steps.
 - True state x_k is within $\hat{x}_k^+ \pm 3\sqrt{\text{diag}(\Sigma_{\hat{x},k}^+)}$ with high confidence.



Summary

- This lesson has reviewed the most important concepts covered in the first three courses of this specialization that relate to particle filters also.
- If you are confident that you understand all the topics mentioned in this review, then you are ready to begin studying particle filters!
- In the next lesson, we will jump right in and discuss the approach that particle filters use to estimate the conditional mean of a nonlinear system’s state, $\mathbb{E}[x_k | \mathbb{Z}_k]$.
- But, we will no longer assume that all pdfs are Gaussian, so we’re going to need to take a much different approach from before, which involves approximating the integral that defines $\mathbb{E}[x_k | \mathbb{Z}_k]$.
 - We end up tracking $f(x_k | \mathbb{Z}_k)$ instead of $\mathbb{E}[x_k | \mathbb{Z}_k]$, which allows us to do much more with the particle-filter result than that of an EKF/SPKF result.
- Without further delay, let’s move to the next lesson and begin!